Dongjun Kim

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RESEARCH INTERESTS

Efficient Edge AI, Algorithm-Hardware Co-Design, Brain-Inspired Neuromorphic Computing

EDUCATION

Kangwon National University — **Gangneung, South Korea**
B.S. Electronic and Semiconductor Engineering (**GPA:** 4.28/4.5, **Rank:** 2/67) — *March 2021 – Feb 2027*
Military Service, Republic of Korea Army(Sergeant) — *Jan 2022 – July 2023*

PUBLICATIONS

- **Quantization Adversarial Distillation for Embedded Underwater Acoustic Target Recognition**
  By **Dongjun Kim**, Sung-Hoon Byun, Sangwook Park*
  Published at IEEE Embedded Systems Letters (ESL), 2026 **[Paper]** **[Code]**

- **Ultra-Lightweight Ship-Radiated Sound Classification for Real-time Embedded Inference**
  By Sangwon Park, **Dongjun Kim**, Sung-Hoon Byun, Sangwook Park*
  Published at IEEE Embedded Systems Letters (ESL), 2026 **[Paper]** **[Code]**

RESEARCH EXPERIENCE

Kangwon National University — *Feb 2025 – Present*
Undergraduate Researcher — *Gangneung, South Korea*

- Advised by **Prof. Sangwook Park** at **Laboratory for Machine Audio Perception(L-MAP)**
- Developed QAD (Knowledge Distillation + Adversarial Training + QAT) for underwater acoustic target recognition on resource-constrained edge devices. Achieved **74.11% accuracy** with **63% model size reduction** and **56% inference latency reduction** on **Raspberry Pi 5**.
- Investigating Sonar-Equation-Based Physics-Informed Diffusion Models to Overcome Data Scarcity in Underwater Acoustic Target Recognition.

Purdue University — *July 2026 – August 2026*
Visiting Undergraduate Researcher — *West Lafayette, USA*

- Advised by **Prof. Martin Byung-Guk Jun** and **Dr. Asif Tanveer**.
- Developed and deployed a real-time Edge AI system for **CNC machine chatter detection** in a real-world manufacturing environment. **[Slides]**
- Received **Outstanding Individual Award** and **Outstanding Project Team Award** (ranked 1st among 21 participants and 7 teams, respectively).

CONFERENCE PRESENTATIONS

- **A Study on Physics-Informed Diffusion for Acoustic Data Augmentation**
  By, Dongjun Kim, Sung-Hoon Byun, Sangwook Park*
  Accepted at IEEE/IEIE International Conference on Consumer Electronics Asia(ICCE-Asia), 2026

- **Adversarial Quantization based Knowledge Distillation for low-complexity vessel sound classification**
  By, Dongjun Kim, Sung-Hoon Byun, Sangwook Park*
  Presented at The Acoustical Society of Korea (ASK), 2026

- **An Approach for Minimizing Processing Error in Sound Event Detection based on BEATs-BiGRU model**
  By Dongjun Kim, Sangwook Park*
  Presented at The Institute of Electronics and Information Engineers (IEIE), 2025

PROJECTS

On-ieum: Integrated Public Transportation AIoT System for the Digitally Vulnerable *Mar 2026 – Present*

- **Role:** Technical Lead (Capstone Project) — Designed end-to-end system pipeline.
- Developed an end-to-end navigation assistance system for digitally vulnerable populations without smartphones, integrating face detection, speech-to-text (STT), an On-device LLM, and public transit APIs.
- Deployed STT, Face Recognition, and Large Language Models on **Edge Devices (Raspberry Pi 5 with AI HAT+, NVIDIA Jetson Orin Nano)** for real-time service. **[Code]**

TECHNICAL SKILLS

- **Programming Languages:** Python, C
- **Frameworks & OS:** PyTorch, Linux, Git, ONNX
- **Hardware Platforms:** Raspberry Pi, NVIDIA Jetson Orin Nano
- **English Proficiency:** TOEFL iBT (80/120, 4.0/6.0)

HONORS & AWARDS

Full-Tuition Merit Scholarship, Kangwon National University	September, 2026
Outstanding Project Team & Outstanding Individual, Purdue-KIAT AI-track Program	August 2026
System-Semiconductor Micro Degree Scholarship, Kangwon National University	March 2026
Outstanding Paper Award, IEIE Conference on Electronics, Semiconductor & AI	August 2025
Academic Excellence Scholarship, Kangwon National University	2024 – 2025
Full-Tuition Merit Scholarship, Kangwon National University	Fall 2021

TEACHING EXPERIENCE

Teaching Assistant, Advanced Programming (C language)	Spring 2026